TITANIC SURVIVAL PREDICTION USING MACHINE LEARNING

Data Science Minor Project

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1. Introduction

- **The Titanic disaster is one of the most significant maritime tragedies in history. Analysing passenger data from this event helps understand the factors that influenced survival during the disaster. This project applies data science and machine learning techniques to predict whether a passenger survived the Titanic accident.**
- **The project demonstrates an end-to-end data science workflow, including data preprocessing, exploratory data analysis, model building, and evaluation using real-world data.**

2. Objective

a) **The main objectives of this project are:**
b) **To analyse the Titanic passenger dataset**
c) **To preprocess and clean the dataset**
d) **To perform exploratory data analysis (EDA)**
e) **To build a machine learning model to predict passenger survival**
f) **To evaluate the model using appropriate performance metrics**

3. Dataset Description

The Titanic dataset contains information about passengers aboard the Titanic. Each record represents a passenger with demographic and travel-related attributes.

Key Features:

- **Passenger Class (P class)**
- **Gender (Sex)**
- **Age**
- **Fare**
- **Port of Embarkation (Embarked)**
- **Target Variable:**
- **Survival status (0 = Not Survived, 1 = Survived)**

The dataset was obtained from an open-source repository and is commonly used for academic and learning purposes.

4. Methodology

✓ **The methodology followed in this project includes the following steps:**
✓ **Data loading and inspection**
✓ **Handling missing values**
✓ **Encoding categorical variables**
✓ **Exploratory Data Analysis (EDA)**
✓ **Feature selection**
✓ **Model training using Logistic Regression**
✓ **Model evaluation using accuracy and confusion matrix**

5. Tools and Technologies Used

- Python
- Pandas and NumPy
- Matplotlib and Seaborn
- Scikit-learn
- Google Collab

6. Implementation Details

- The implementation was carried out in Google Collab using Python. Missing values in the Age column were handled using median imputation, while missing values in the Embarked column were filled using the most frequent category. The Cabin column was removed due to a high number of missing values.
- Categorical variables such as gender and embarkation port were encoded into numerical values. Logistic Regression was used as the classification algorithm due to its simplicity and effectiveness in binary classification problems.

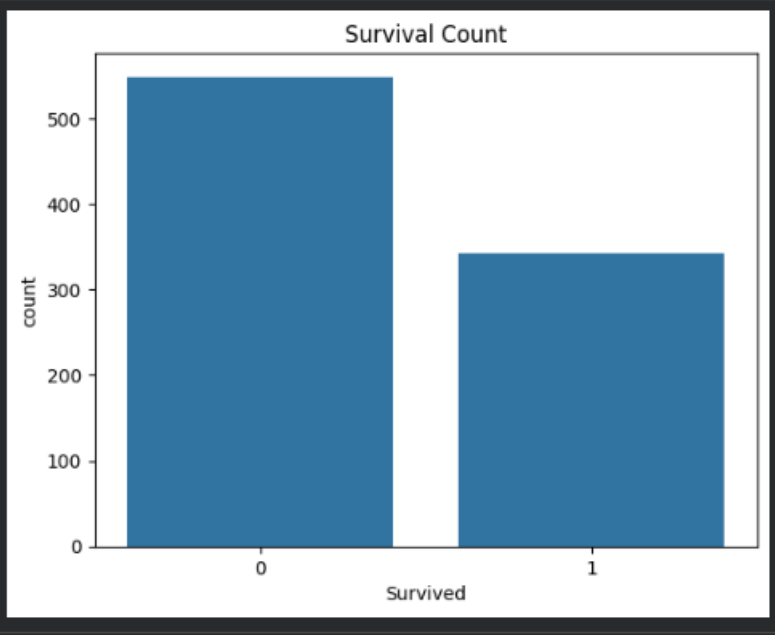
7. Results and Observations

The results obtained from exploratory data analysis and model evaluation are presented below.

❖ Figure 1: Survival count of passengers

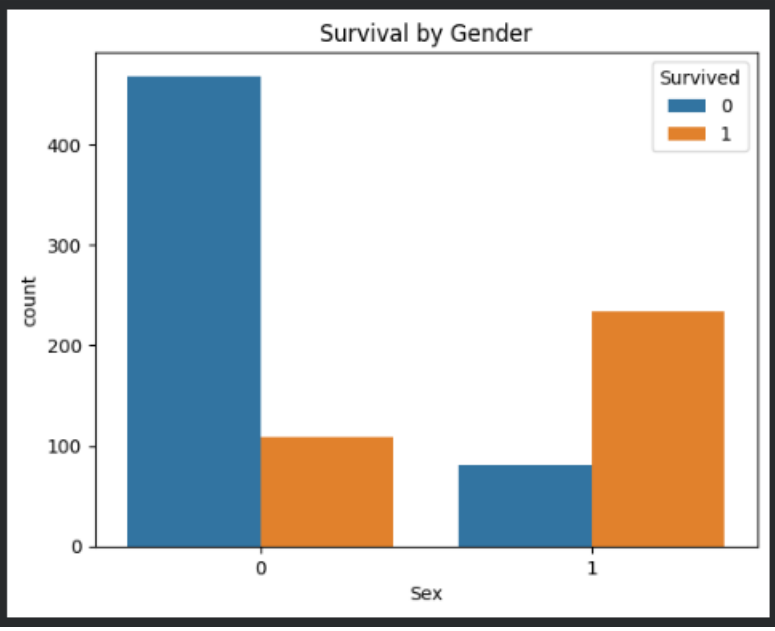
```
df['Sex'] = df['Sex'].map({'male': 0, 'female': 1})
df['Embarked'] = df['Embarked'].map({'S': 0, 'C': 1, 'Q': 2})
```

```
sns.countplot(x='Survived', data=df)
plt.title("Survival Count")
plt.show()
```



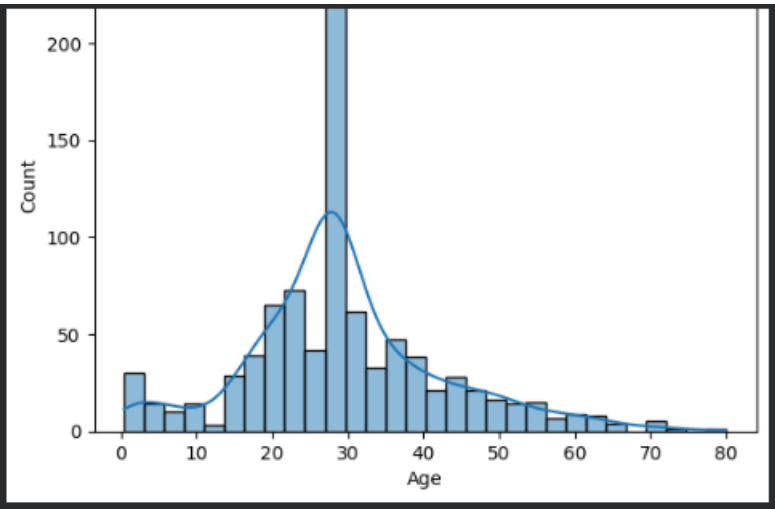
❖ **Figure 2: Survival comparison by gender**

```
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title("Survival by Gender")
plt.show()
```



❖ **The above visualizations indicate that female passengers had a higher survival rate compared to male passengers.**

Figure 3: Model accuracy output




```
X = df[['Pclass', 'Sex', 'Age', 'Fare', 'Embarked']]
y = df['Survived']
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

Dataset was split into 80% training and 20% testing data.

```
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

Show hidden output

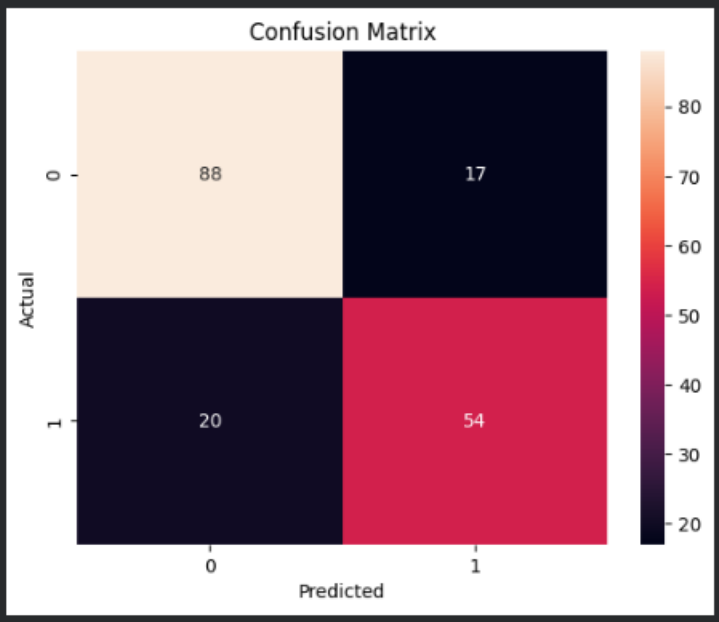
```
y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

```
Accuracy: 0.7932960893854749
```

- **The Logistic Regression model achieved satisfactory accuracy, indicating good predictive performance.**

Figure 4: Confusion matrix of Logistic Regression model

```
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt='d')
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```




```
print(classification_report(y_test, y_pred))
```

```
              precision    recall  f1-score   support

           0       0.81      0.84      0.83       105
           1       0.76      0.73      0.74        74

    accuracy                           0.79       179
   macro avg       0.79      0.78      0.79       179
weighted avg       0.79      0.79      0.79       179
```

- **The confusion matrix shows that the model correctly classified most passengers, with some misclassifications.**

8. Conclusion

- **This project successfully demonstrates the application of data science and machine learning techniques to predict passenger survival using the Titanic dataset. Through data preprocessing, visualization, and model training, meaningful insights were obtained regarding the factors affecting survival.**
- **The Logistic Regression model provided reliable predictions and highlighted the importance of gender and passenger class in survival outcomes. This project serves as a practical example of how machine learning can be applied to real-world classification problems.**